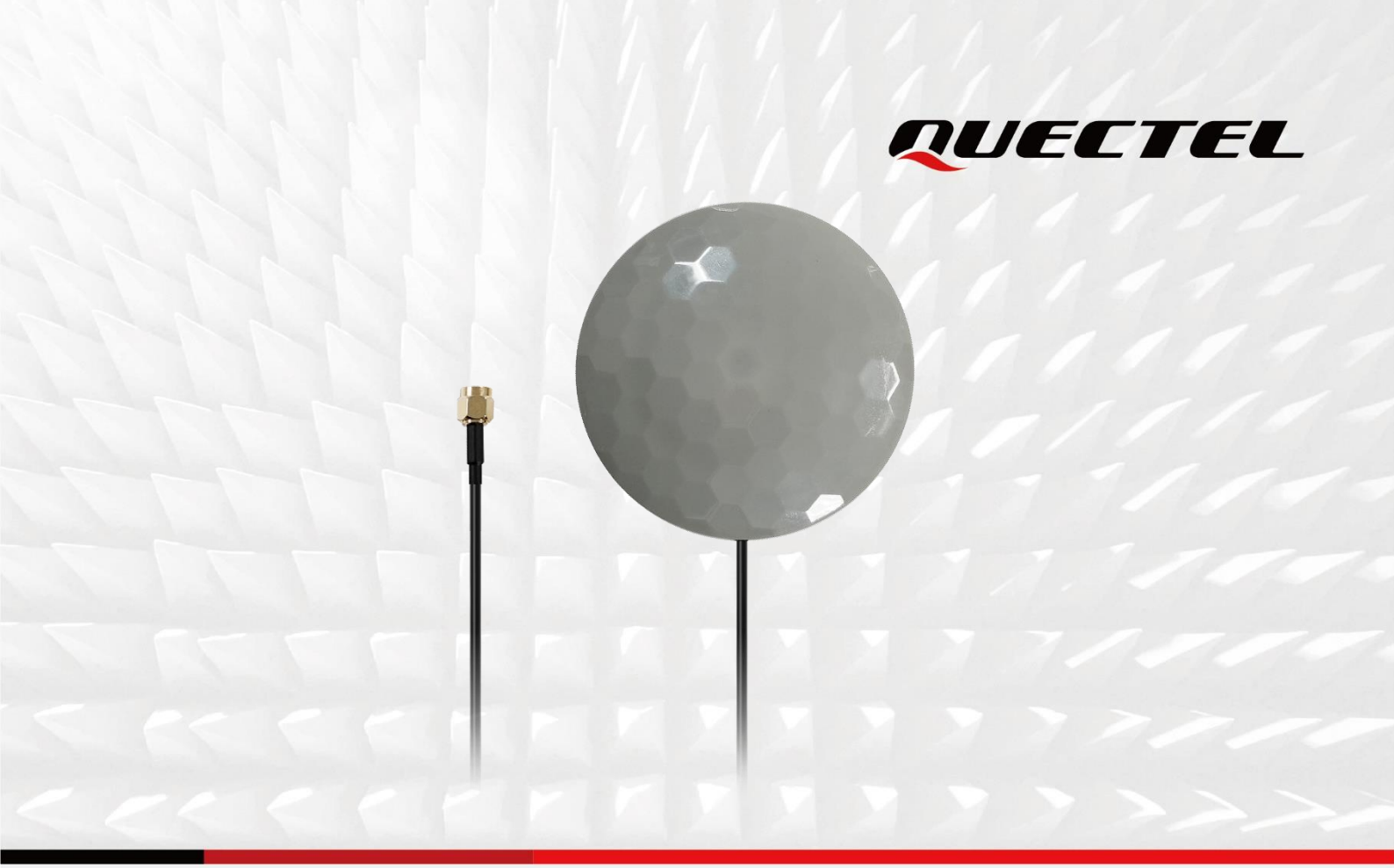


# Antenna Datasheet

**Product OC:** YECA000M1A
**Version:** 1.7
**Date:** 2026-03-23
**Status:** Released

**Product Name:** 4G Adhesive Mount Low Profile Monopole External Antenna

**Key Features:**

Frequency Band: 698–960 MHz, 1710–2690 MHz
Dimensions: Φ 99.98 mm × 17.46 mm
Efficiency: Up to 52 % (FS)
RoHS and REACH Compliant
IP68, IP69K, IK10
Pressure load: 1500 lbs MAX

# Overview

The YECA000M1A antenna is a self-developed Quectel design that delivers exceptional performance and stability, making it suitable for the challenging environments found in municipal road applications. This antenna features a paste-on installation method, allowing for easy attachment to roads, sidewalks, or manholes. Its robust, non-reflective, and low-profile design ensures consistent and reliable operation in diverse settings.

Operating within the 4G LTE frequency bands, the YECA000M1A provides high-speed and stable signal transmission. It comes with an integrated SMA-M direct mount connector for seamless connectivity with other communication devices. To meet customer demands, the YECA000M1A 4G antenna also offers customizable cable and connector options.

The YECA000M1A 4G antenna is instrumental in smart city initiatives and is widely applied in intelligent transportation, monitoring, and security systems, making it an ideal choice for 4G communications. This weather-resistant, paste-on marking antenna ensures a stable and reliable wireless connection in various environments. For installation ease, it is recommended to use durable adhesive solutions like 3M VHB 4955 to securely attach the antenna to its intended location.

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# 1 Specification

**Test Condition: In Free Space**

## 1.1. Electrical

| Electrical | |
|---|---|
| **Frequency Range** | 698–960 MHz, 1710–2690 MHz |
| **Impedance** | 50 Ω |
| **Polarization** | Linear |
| **Radiation Pattern** | Omni-directional |

| Electrical – Detail | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Band / SPEC | Band | B71 | B12 /B13 /B28 | B5 /B8 /B26 | n74 /n75 /n76 | B1 /B2 /B3 | B40 | Wi-Fi 2G | B38 /B41 | B42 /B48 /n77 | n79 | Wi-Fi 5G |
| | Freq. (MHz) | 600–700 | 700–810 | 820–960 | 1420–1520 | 1700–2170 | 2300–2400 | 2400–2500 | 2500–2690 | 3300–4200 | 4400–5000 | 5150–5850 |
| Max. VSWR | FS | - | 2.1 | 4.2 | - | 2.6 | 2.9 | 2.7 | 2.8 | - | - | - |
| Max. Return Loss (dB) | FS | - | -9.1 | -4.2 | - | -7.0 | -6.3 | -6.7 | -6.5 | - | - | - |
| AVG Eff. (%) | FS | - | 49.4 | 31.7 | - | 33.7 | 27.8 | 28.5 | 26.1 | - | - | - |
| AVG. AVG Gain (dB) | FS | - | -3.1 | -5.3 | - | -4.7 | -5.6 | -5.5 | -5.8 | - | - | - |
| Max. Peak Gain (dBi) | FS | - | 2.9 | 2.6 | - | 1.8 | -1.0 | -0.7 | -1.5 | - | - | - |

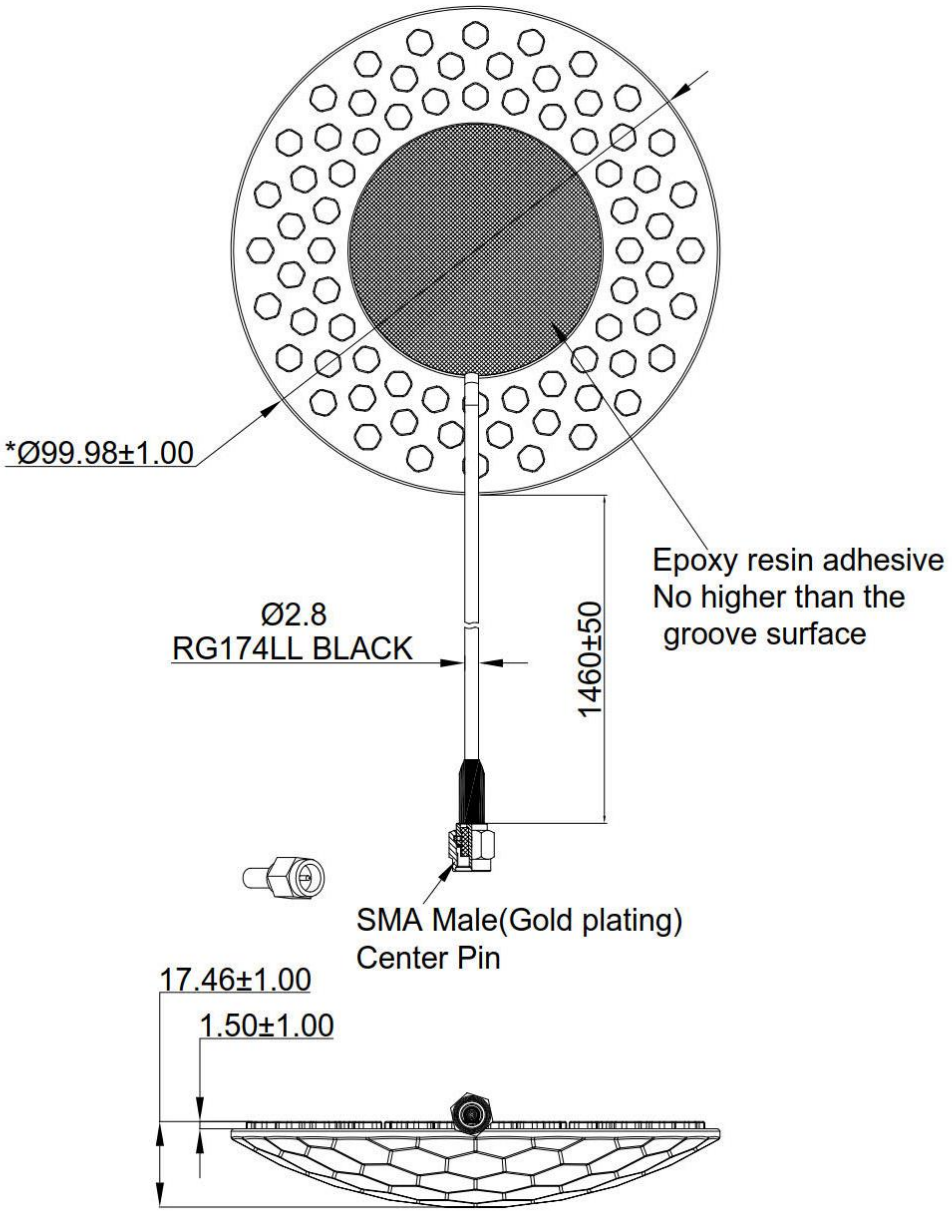
| | | |
|---|---|---|
| VSWR | FS | ≤ 4.2 |
| Return Loss | FS | ≤ -4.2 dB |
| Peak Gain | FS | ≤ 2.9 dBi |

- FS: In Free Space

## 1.2. Mechanical & Environmental

| | |
|---|---|
| **Mechanical** | |
| **Antenna Dimensions** | Φ 99.98 mm × 17.46 mm |
| **Casing Material & Color** | ASA & Grey |
| **Cable Type & Color & Length** | RG174LL & Black & 1460 mm |
| **Connector Type** | SMA Male |
| **Mounting Type** | Adhesive |
| **Weight** | Typ. 101.5 g |
| **Environmental** | |
| **Operation Temperature** | -40 °C to +85 °C |
| **Storage Temperature** | -40 °C to +85 °C |
| **Ingress Protection (IP) Rating** | IP68 (water depth 2 m for 1 hour), IP69K |
| **Impact Protection (IK) Rating** | IK10 |
| **RoHS & REACH Compliant** | Yes |
| **Housing UV Resistant** | UL 746c f1 |

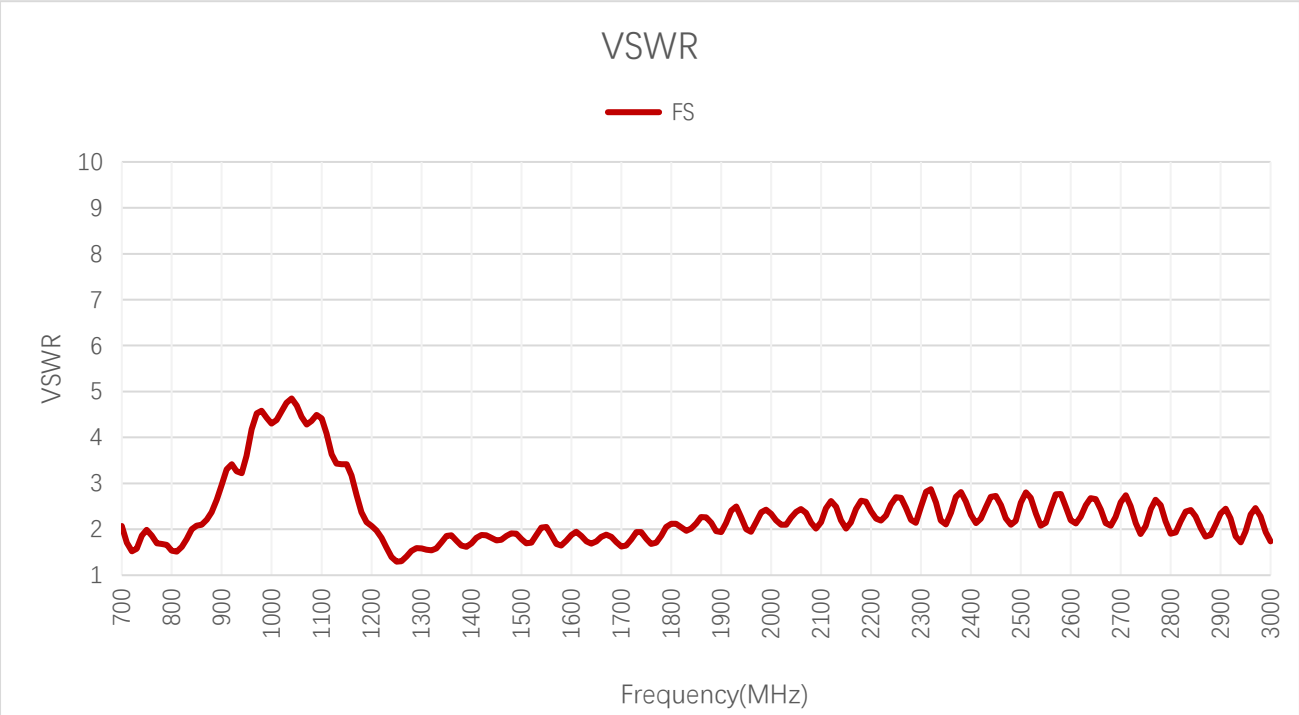
# 2 Drawing


*Ø99.98±1.00
Epoxy resin adhesive
No higher than the
groove surface
Ø2.8
RG174LL BLACK
1460±50
SMA Male(Gold plating)
Center Pin
17.46±1.00
1.50±1.00

# 3 Detailed Performance

## 3.1. S-Parameter Test

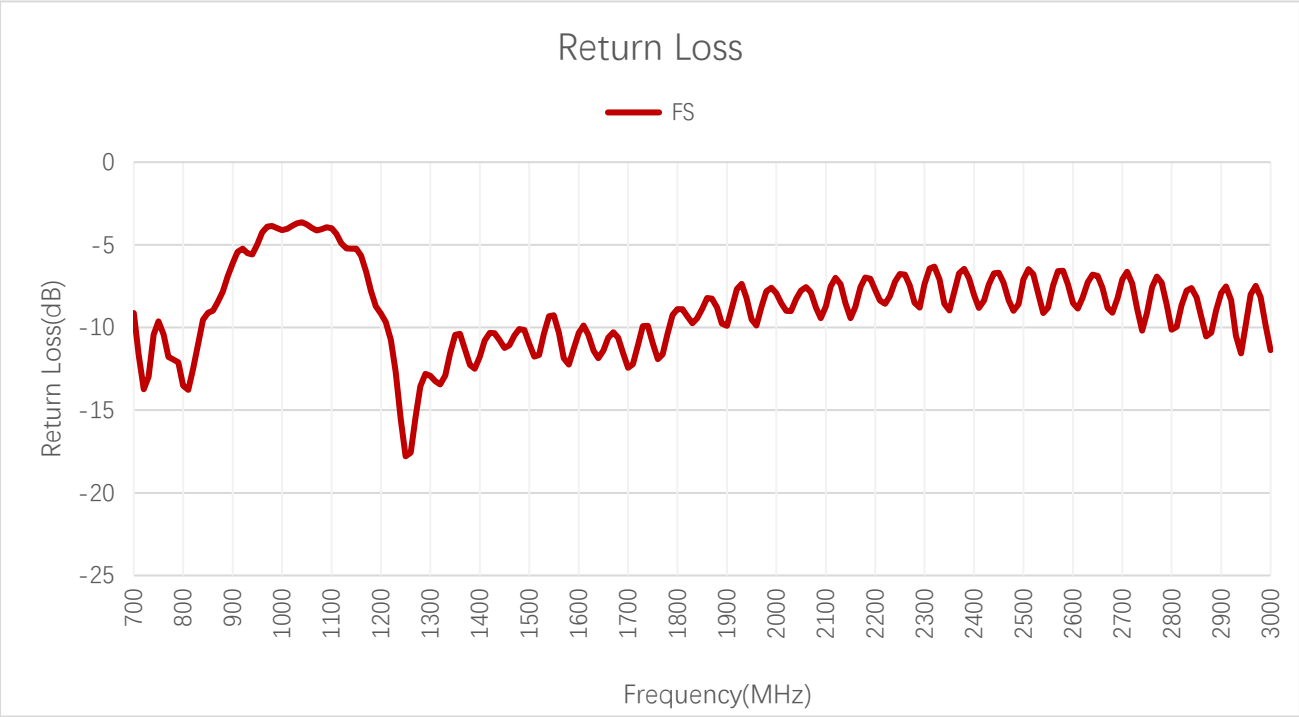
### 3.1.1. VSWR




**VSWR**

| Frequency (MHz) | 600 | 630 | 710 | 830 | 900 | 960 | 1440 | 1710 | 1740 | 1880 |
|---|---|---|---|---|---|---|---|---|---|---|
| VSWR | - | - | 1.7 | 1.8 | 3.0 | 4.2 | - | 1.6 | 1.9 | 2.1 |
| Frequency (MHz) | 1950 | 2140 | 2350 | 2450 | 2600 | 2690 | 4700 | 5000 | 5500 | 6000 |
| VSWR | 2.0 | 2.2 | 2.1 | 2.7 | 2.2 | - | - | - | - | - |

### 3.1.2. Return Loss

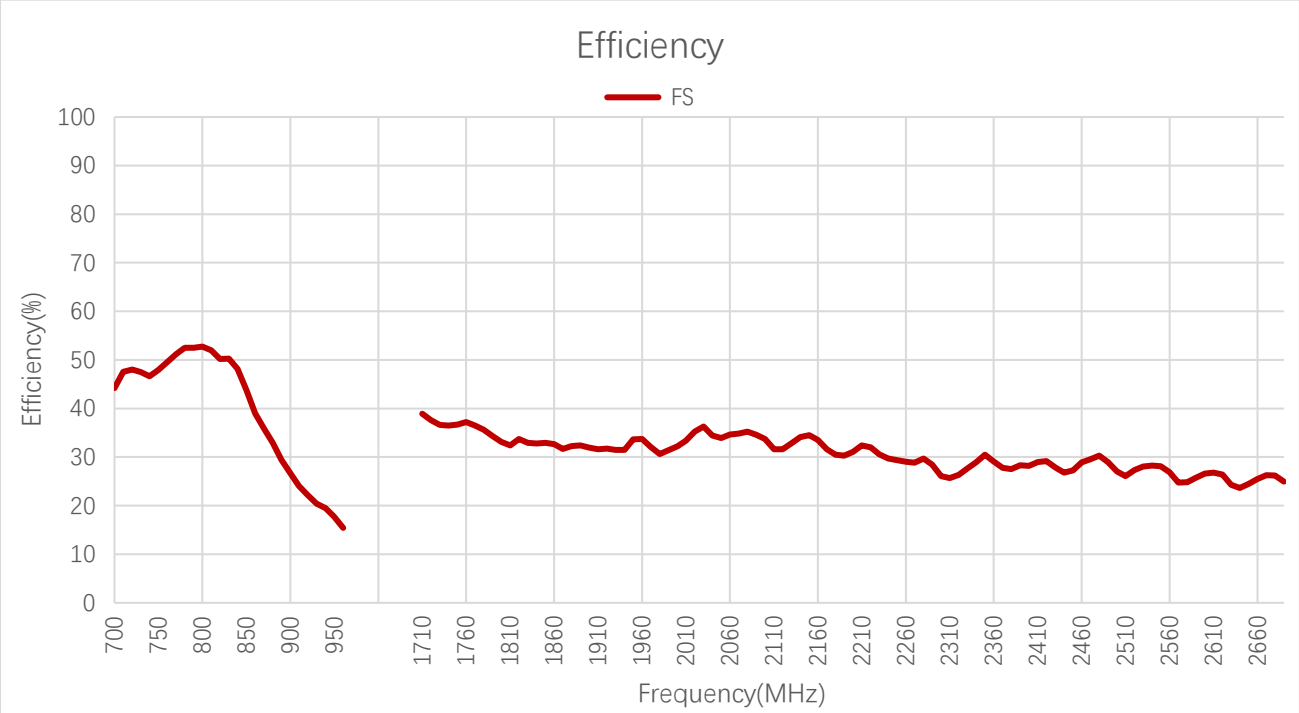



**Return Loss (dB)**

| **Frequency (MHz)** | **600** | **630** | **710** | **830** | **900** | **960** | **1440** | **1710** | **1740** | **1880** |
|---|---|---|---|---|---|---|---|---|---|---|
| **Return Loss (dB)** | - | - | -11.6 | -11.0 | -6.1 | -4.2 | - | -12.2 | -9.9 | -8.8 |
| **Frequency (MHz)** | **1950** | **2140** | **2350** | **2450** | **2600** | **2690** | **4700** | **5000** | **5500** | **6000** |
| **Return Loss (dB)** | -9.5 | -8.5 | -9.0 | -6.7 | -8.5 | - | - | - | - | - |

## 3.2. Radiation Performance Test

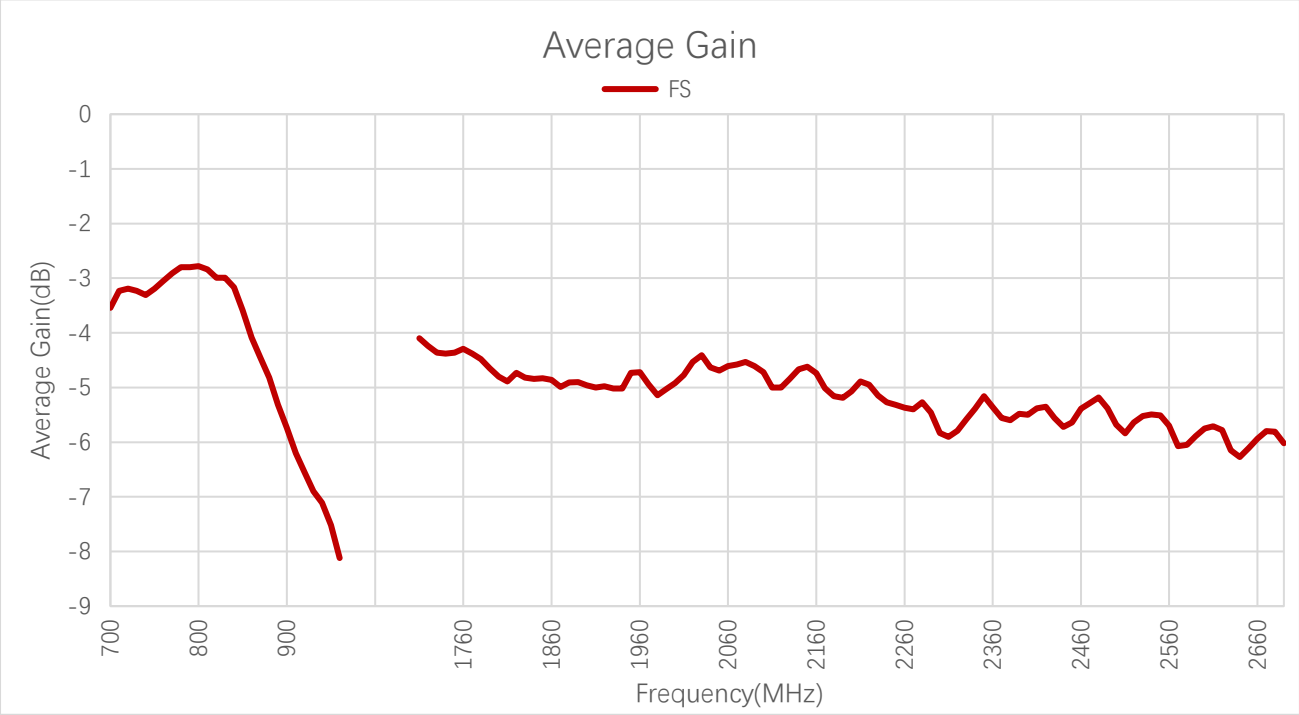
### 3.2.1. Efficiency




**Efficiency (%)**

| **Frequency (MHz)** | **600** | **630** | **710** | **830** | **900** | **960** | **1440** | **1710** | **1740** | **1880** |
|---|---|---|---|---|---|---|---|---|---|---|
| **Efficiency (%)** | - | - | 47.6 | 50.3 | 26.7 | 15.4 | - | 38.9 | 36.5 | 32.3 |
| **Frequency (MHz)** | **1950** | **2140** | **2350** | **2450** | **2600** | **2690** | **4700** | **5000** | **5500** | **6000** |
| **Efficiency (%)** | 33.6 | 34.1 | 30.5 | 27.3 | 26.6 | - | - | - | - | - |

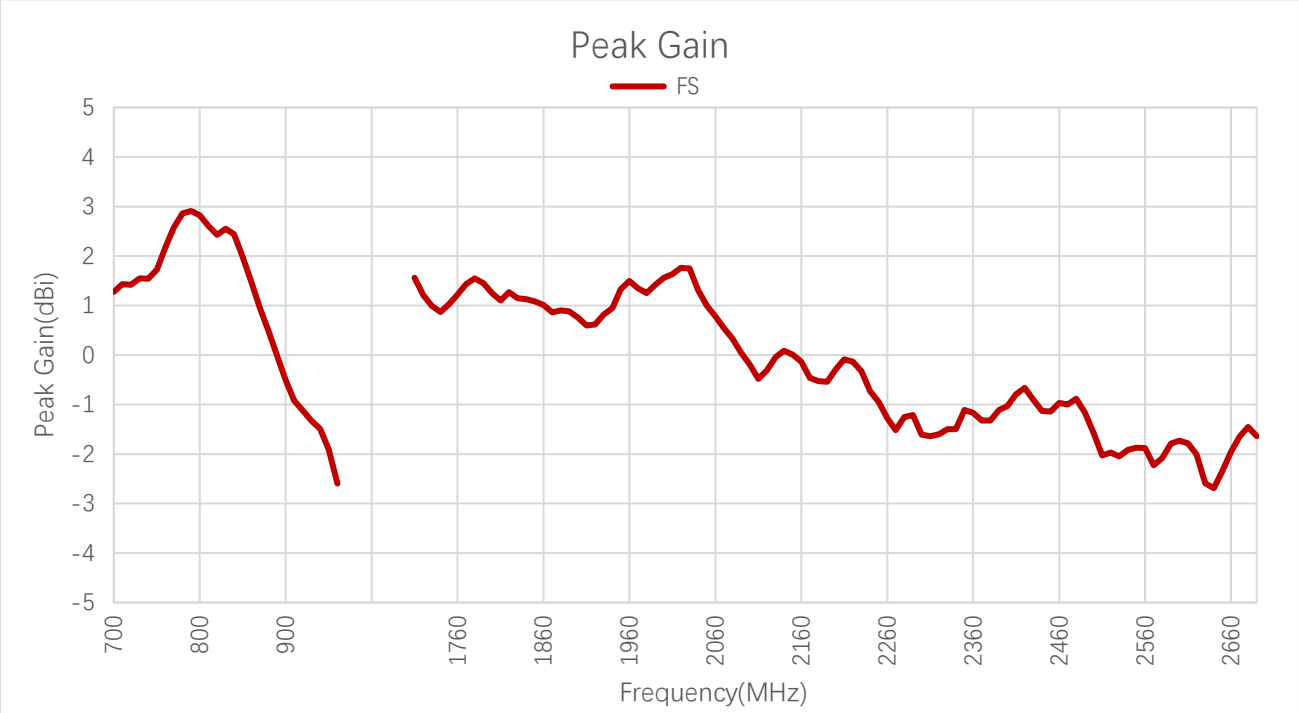
### 3.2.2. Average Gain




**Average Gain (dB)**

| **Frequency (MHz)** | **600** | **630** | **710** | **830** | **900** | **960** | **1440** | **1710** | **1740** | **1880** |
|---|---|---|---|---|---|---|---|---|---|---|
| **Average Gain (dB)** | - | - | -3.2 | -3.0 | -5.7 | -8.1 | - | -4.1 | -4.4 | -4.9 |
| **Frequency (MHz)** | **1950** | **2140** | **2350** | **2450** | **2600** | **2690** | **4700** | **5000** | **5500** | **6000** |
| **Average Gain (dB)** | -4.7 | -4.7 | -5.2 | -5.6 | -5.8 | - | - | - | - | - |

### 3.2.3. Peak Gain



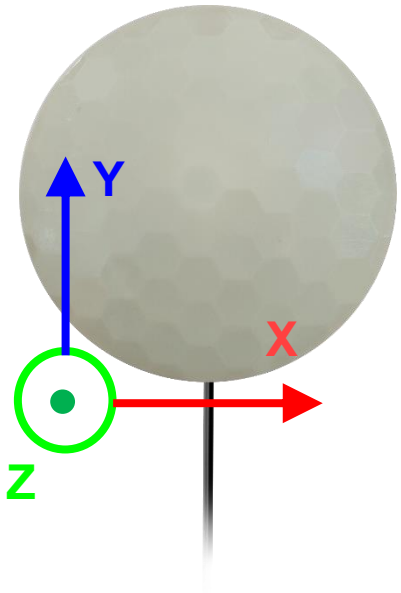

**Peak Gain (dBi)**

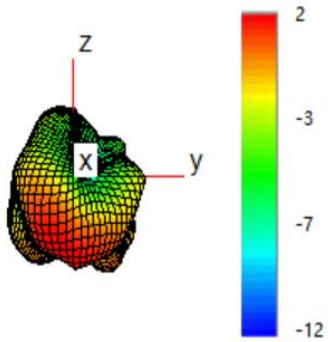
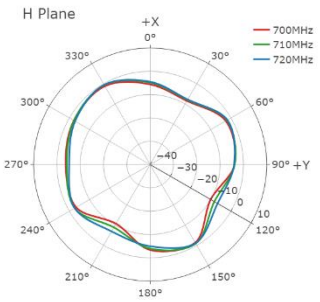
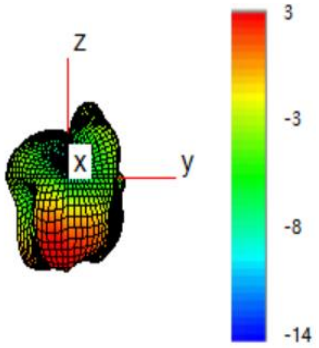
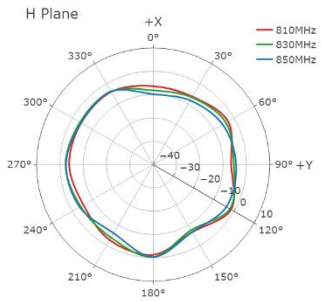
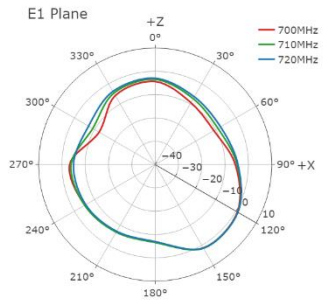
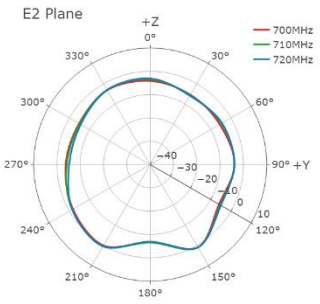
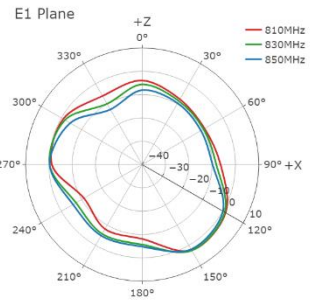
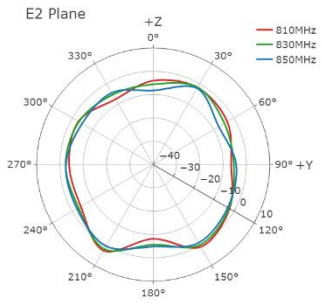
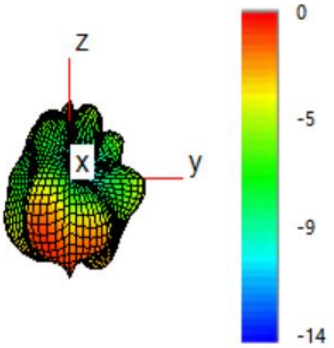
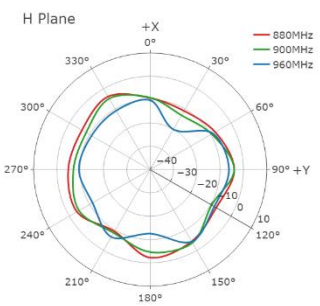
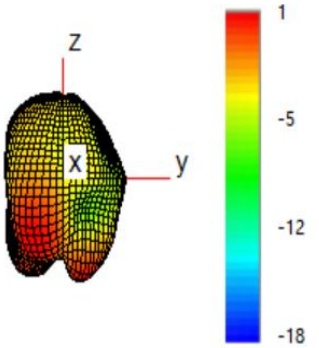
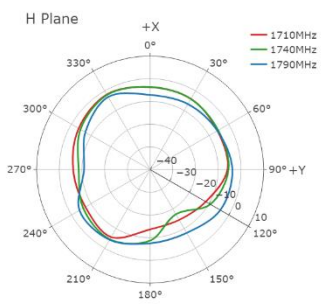
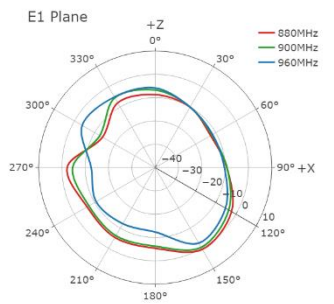
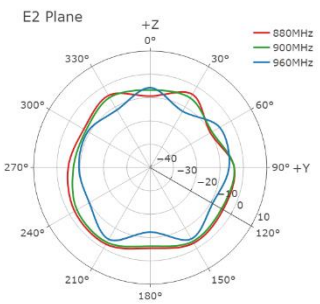
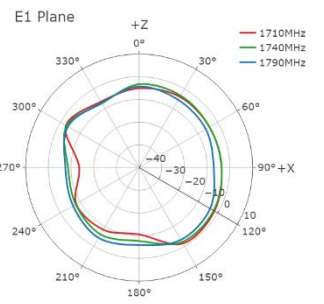
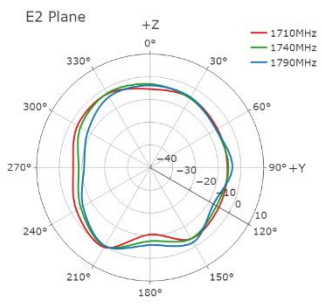
| **Frequency (MHz)** | **600** | **630** | **710** | **830** | **900** | **960** | **1440** | **1710** | **1740** | **1880** |
|---|---|---|---|---|---|---|---|---|---|---|
| **Peak Gain (dBi)** | - | - | 1.4 | 2.6 | -0.5 | -2.6 | - | 1.6 | 0.9 | 0.9 |
| **Frequency (MHz)** | **1950** | **2140** | **2350** | **2450** | **2600** | **2690** | **4700** | **5000** | **5500** | **6000** |
| **Peak Gain (dBi)** | 1.3 | 0.1 | -1.1 | -1.1 | -1.7 | - | - | - | - | - |

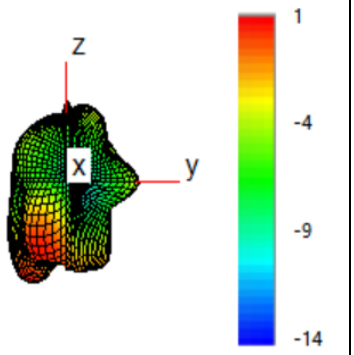
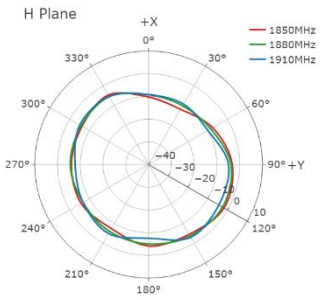
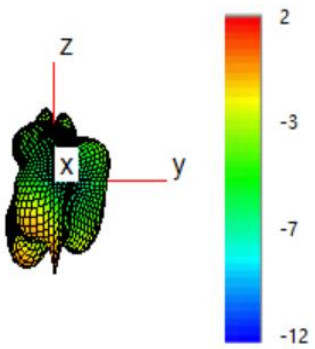
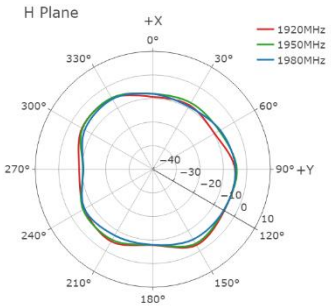
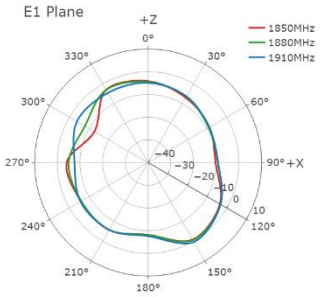
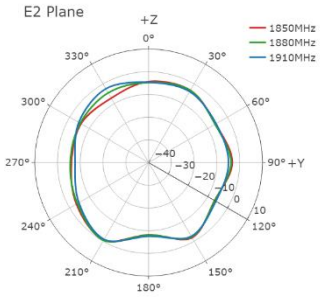
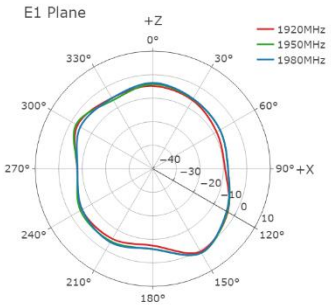
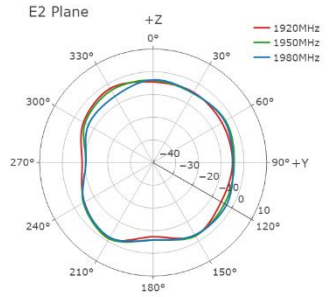
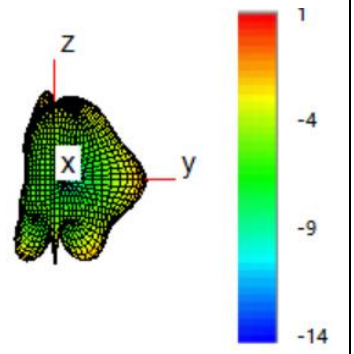
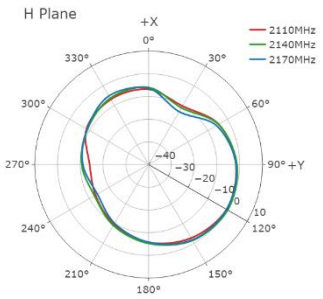
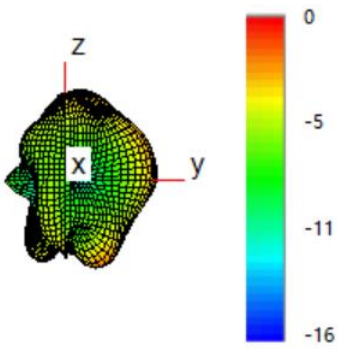
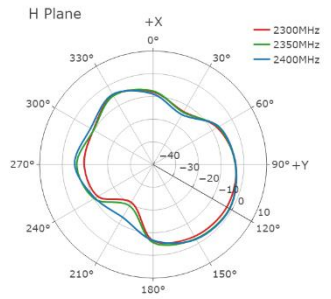
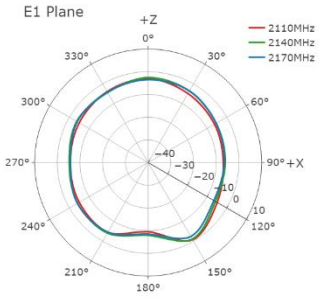
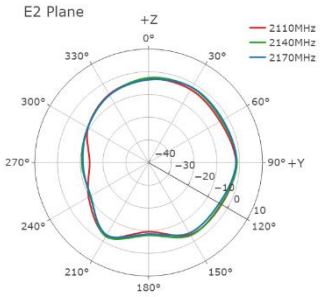
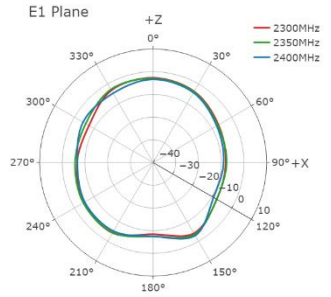
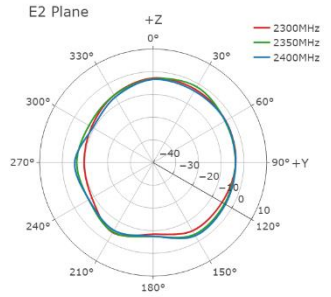
### 3.2.4. 3D & 2D Radiation Pattern

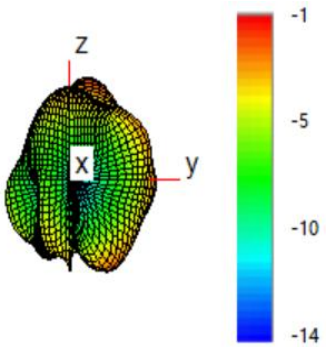
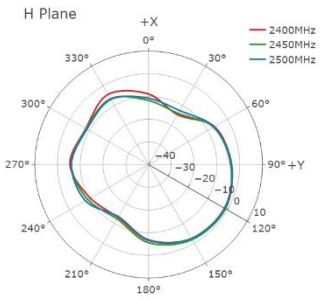
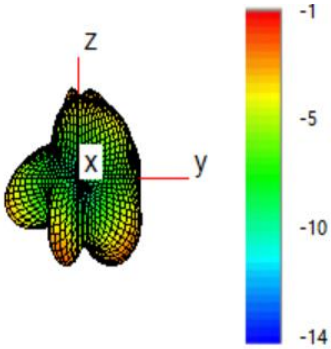
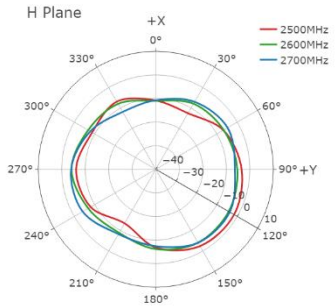
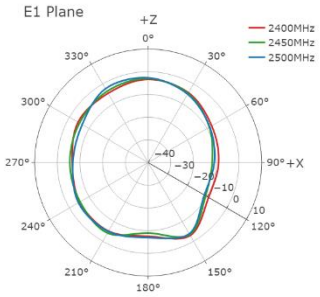
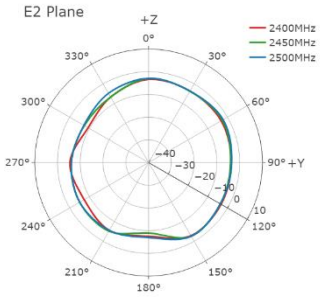
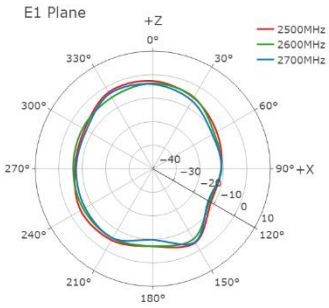
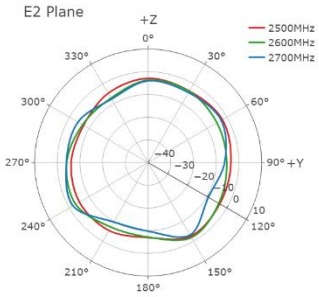
#### 3.2.4.1. Test Condition: Free Space

- Test Chamber: GL-S-1

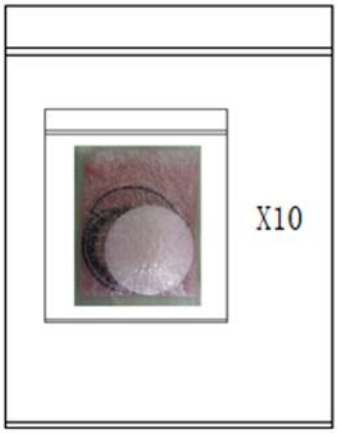
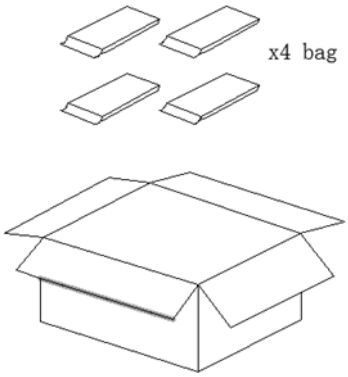


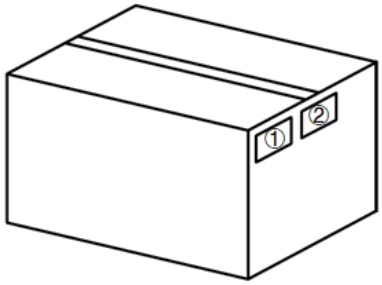
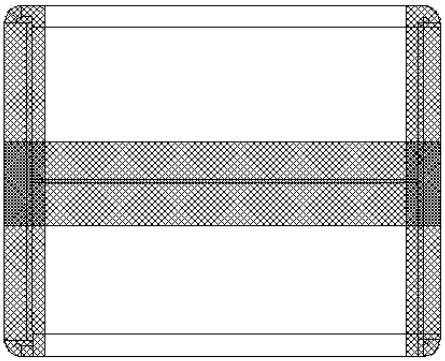
| 710 MHz | | 830 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |
| 900 MHz | | 1740 MHz | |
|  |  |  |  |
|  |  |  |  |

| 1880 MHz | | 1950 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |
| 2140 MHz | | 2350 MHz | |
|  |  |  |  |
|  |  |  |  |

| 2450 MHz | | 2600 MHz | |
|---|---|---|---|
|  |  |  |  |
|  |  |  |  |

# 4 Packaging

| Step | Packaging Picture / 2D Picture | Description |
|---|---|---|
| 1 |  | After covering the bubble bag, the product is put into the PE bag.<br>1 antenna product in a PE bag.<br>(1 Antenna / PE Bag) |
| 2 |  | 10 PE bags in a big PE bag.<br>(10 Antennas / Big PE Bag) |
| 3 |  | (4 Big PE Bags / Carton Box)<br>(40 Antennas / Carton Box)<br>Carton Size:<br>L × W × H = 405 × 293 × 185 mm |

| | | |
|---|---|---|
| 4 |  | **Position for Attaching Labels**<br>① Carton Label<br>② Quality Label |
| 5 |  | **Sealing Cartons**<br>H-shaped sealing cartons |
| Note | The initial packaging method described above is for reference only, and the final actual packaging method shall be subject to the actual shipping packaging. | |

# Contact Us

**At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:**

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**Or our local offices. For more information, please visit:**
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# Revision History

| Version | Date | Author | Note |
|---|---|---|---|
| - | 2023-07-10 | Andy Yan/ Toby Wang/ David Liu/ Vinnie Liu | Creation of the document |
| 1.0 | 2023-07-10 | Andy Yan/ Toby Wang/ David Liu/ Vinnie Liu | First official release |
| 1.1 | 2023-12-05 | Toby Wang/ Vinnie Liu | Updated casing color (Chapter 1.2). |
| 1.2 | 2024-01-22 | Toby Wang | Updated cable type to ALSR 100 (Chapters 1.2 and 2). |
| 1.3 | 2024-11-02 | Toby Wang | Added housing UV resistant (Chapter 1.2). |
| 1.4 | 2025-01-09 | Toby Wang | Updated the drawing (Chapter 2). |
| 1.5 | 2025-03-11 | Rainey Liao | Updated the starting frequency to 698 MHz (Homepage and Chapter 1.1) |
| 1.6 | 2025-04-24 | Rainey Liao | Updated the antenna image (Cover page). |
| 1.7 | 2026-03-23 | Toby Wang/ Rainey Liao | 1. Updated the antenna image (Cover page).<br>2. Updated cable type to RG174LL (Chapters 1.2 and 2). |



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